An Exact Counterexample to Graffiti 322

AdaIvy project*

Lean verification. Passed for one linked formal artifact. A linked artifact can verify only part of a claim. AdaIvy labels a claim a Theorem only when its complete representation is kernel checked with no unapproved assumptions.

Claim status: this report contains no fully Lean-verified Theorems, one exact-certificate Proposition, and no unresolved or conjectural Proposals.

Environment names are computed from the records that back each claim; no input field selects one, and every missing record weakens the rendered claim rather than strengthening it.

Novelty: `not_assessed`. Significance: `not_assessed`. Neither is inferred from any proof status, and no automated assessment of either exists in this system.

Prior-result classification: `not_assessed`. No pre-announcement classification record is attached.

Corpus provenance: `project_authored`. No human publication approval record exists. Automated checks alone do not make this an endorsed AdaIvy result, so this document is a rendered projection and not an approved publication.

Formal backend pinned at Lean `v4.32.1` commit `f054605aea4b`, mathlib `v4.32.1` commit `520045ab14e2`. Every solved claim has a linked `.lean` artifact. File creation is not formal verification; each link states `pending`, `failed`, or `kernel_checked` from its record.

Abstract

We give an exact counterexample to the source-faithful formulation of Graffiti 322. A connected triangle-free graph on 448 vertices has Inverse Even 40049/4444, strictly greater than its nine distinct distance eigenvalues. Two exact checkers verify the construction and spectral decomposition; a sealed Lean check covers the closed arithmetic certificate. Novelty and significance remain not assessed.

Run disclosure

Recorded model activity: openai / `gpt-5-mini-2025-08-07` (5 completed call(s)); azure_openai / `gpt-5.6-sol` (0 completed call(s)). Recorded model calls: 5. Recorded external model cost: USD 0.029632 (authorized cap USD 40). Recorded tokens: 13114 input, 13175 output, 26289 total. Measurement status: `partial`. Provider telemetry covers the bounded AdaIvy sessions only. The surrounding interactive Codex authoring session exposed no billable usage to the ledger and is neither included nor estimated.

*AI-assisted mathematical research system

1 Statement and conventions

Graffiti 322 concerns finite simple connected triangle-free graphs. Even(v) counts vertices at even graph distance from v, including v itself because distance zero is even. Range is read as the number of distinct distance eigenvalues, consistently with the source's separate use of range and scope. [1]

Proposition 1 (record `cl.graffiti-322-counterexample`)**.** *The connected triangle-free graph G(14,18) has Inverse Even 40049/4444 and exactly nine distinct distance eigenvalues, so it refutes Graffiti 322 under the frozen source-faithful conventions.*

$$\mathrm{InvEven}(G_{14,18}) = \frac{40049}{4444} > 9 = |\mathrm{spec}(D(G_{14,18}))|$$

Provenance. *Evidence class* `exact_certificate_proposition`*: an exact certificate separates the two values by 53/4444 in fractions-exact arithmetic with no floating point. Records:* `att.graffiti-322-arithmetic`*,* `cert.graffiti-322-exact-separation`*,* `cite.roucairol-cazenave.graffiti-322`*,* `cite.wow.even-definition`*,* `cl.graffiti-322-counterexample`*. Representation* `rep.graffiti-322-partial-formalization.v1` *is* `partially_verified`*. The certificate is offered as* `separates_values`*. Background: [1]; [2]. Original problem: [1].*

Contribution. This result was AI-generated by AdaIvy project.

Auditable derivation. Construct G(r,t) by replacing every edge of K(r) with a length-three path and attaching t leaves at each branch vertex. Classify the three even-distance counts exactly. Decompose the full distance representation into scalar -2 and -1 spaces and two quotient actions of dimensions three and four. At r=14 and t=18 the dimensions sum to 448, the quotient polynomials contribute at most seven further eigenvalues, and exact rational arithmetic gives Inverse Even 9 + 53/4444. Attribution and source influence: [1].

Formal artifact. status `kernel_checked`; `lean/AdaIvyPhase3B_AdaIvyGraffiti322ArithmeticCertificate.lean`

Checked Lean statement (`att.graffiti-322-arithmetic`):

```
: (14 * 14 + 14 * 18 : Nat) = 448 ∧ (3 * 14 * 13 / 2 + 14 * 18 : Nat) = 525 ∧ (1
+ 13 * 13 + 13 * 18 : Nat) = 404 ∧ (3 * 14 - 4 + 18 : Nat) = 56 ∧ (2 * 14 - 2 + 18
: Nat) = 44 ∧ (3 + 4 * 13 + (14 * 14 - 3 * 14 + 1) + 14 * 17 : Nat) = 448 ∧ (9 *
4444 : Nat) = 39996 ∧ (40049 - 39996 : Nat) = 53 ∧ ¬ ((40049 : Nat) ≤ 9 * 4444)
```

Case	`graffiti-322-g14-18`
Run	`run.graffiti-322.exact-family.v1`
Engine	`exact_graph_distance_and_invariant_space_v2`
Arithmetic	`fractions-exact`
Floating point used	`no`
Primal value	`40049/4444`
Dual value	`9`
Duality gap	`53/4444`
Classification	`counterexample_values_separated`
Engine result hash	`sha256:2e93bd5ac6829079f89091300d0dad0146f08e5e2354f973 4b6384874c6edb6c`

2 Construction and exact Inverse Even value

For r at least four and t at least two, take branch vertices b_i, oriented internal vertices x_ij for i not equal to j, and leaves l_i,a. Add exactly the edges b_i x_ij, x_ij x_ji, and b_i l_i,a. Thus each edge of K_r becomes a path of length three and t leaves attach to every branch. The graph is connected and triangle-free.

$$|V(G_{r,t})| = r^2 + rt, \qquad |E(G_{r,t})| = \frac{3r(r-1)}{2} + rt$$

Direct distance classification gives three even-distance counts: one for branch vertices, one for oriented internal vertices, and one for leaves.

$$E(b_i) = 1 + (r-1)^2 + (r-1)t, \quad E(x_{ij}) = 3r - 4 + t, \quad E(l_{i,a}) = 2r - 2 + t$$

$$\mathrm{InvEven}(G_{14,18}) = \frac{14}{404} + \frac{182}{56} + \frac{252}{44} = \frac{40049}{4444} = 9 + \frac{53}{4444}$$

3 Exact distance-spectrum decomposition

The distance operator preserves four components: within-cluster leaf differences of dimension 238 with eigenvalue -2; off-diagonal internal arrays with zero row and column sums of dimension 155 with eigenvalue -1; a three-dimensional orbit-constant space; and thirteen copies of one four-dimensional standard space. Their dimensions sum to 448, so the decomposition spans the full vertex space.

$$238 + 155 + 3 + 4 \cdot 13 = 448$$

$$Q_0 = \begin{pmatrix} 39 & 663 & 954 \\ 51 & 805 & 1170 \\ 53 & 845 & 1204 \end{pmatrix}$$

Orbit-constant block.

$$Q_1 = \begin{pmatrix} -3 & -37 & -23 & -54 \\ -3 & -38 & -23 & -54 \\ -2 & -25 & -12 & -36 \\ -3 & -37 & -23 & -56 \end{pmatrix}$$

Standard block, repeated thirteen times.

$$\det(\lambda I - Q_0) = \lambda^3 - 2048\lambda^2 - 25454\lambda - 54228,$$
$$\det(\lambda I - Q_1) = \lambda^4 + 109\lambda^3 - 146\lambda^2 - 442\lambda - 20.$$

Together with the scalar eigenvalues -1 and -2, the two quotient blocks give at most nine distinct eigenvalues. An exact square-free calculation gives gcd(P,P') equal to one for their product with the two scalar factors, so the count is exactly nine. The upper bound alone already yields the counterexample.

4 Verification, literature, and scope

The principal exact checker constructs all 448 vertices, computes every graph distance, verifies triangle-freeness and the invariant action on exact spanning bases, and derives the rational characteristic polynomials. A separate checker recomputes the formulas and determinants independently. The sealed Lean finding checks only the closed arithmetic and dimension identities, with no axioms; it does not formalize the complete graph-distance or invariant-space argument.

The pre-research review inspected the primary source, the Roucairol-Cazenave preprint and publication record, and exact-phrase searches for the conjecture and its wording. It found no independent later proof or counterexample under that bounded protocol. Because this search predates the present construction, a result-specific before-announcement re-check remains open. The correct current label is candidate refutation; novelty and significance are not assessed. [2]

Appendix A. Axiom dependence

Each row is the recorded output of a kernel check plus its `#print axioms` result, against the pinned toolchain named in the status block.

Declaration	`AdaIvyPhase3B.AdaIvyGraffiti322ArithmeticCertificate`
Outcome	`kernel_checked`
Approved axioms	none
Unapproved assumptions	none

Appendix B. Open obligations

Background this document relies on but holds no record for appears here rather than in the bibliography. An unrecorded reference is an open obligation, not a citation.

`obl.graffiti-322-before-announcement-novelty-recheck` Run the result-specific before-announcement novelty re-check before any publication approval. Status `open`. The pre-research scoped search predates the G(14,18) construction and cannot establish priority for it.

`obl.graffiti-322-human-graph-proof-review` A human reviewer must check the graph-distance classification and invariant-subspace decomposition. Status `open`. The sealed Lean artifact covers closed arithmetic, not the complete 448-vertex graph and spectrum derivation.

Appendix C. Reproduction

This document is a projection of manuscript `ms.graffiti-322-counterexample.v1` with content hash `sha256:847b01a7b1c69811860cc0ee987c8794dca714dc3ee0207ec222f0b3743d9e9c`, rendered against template `sha256:e58db90ca47d8d2be716af8ce4a1a0bba328ac56bdbbc338d7a1d4f1d8bfcca8`. The bundle accompanying this document carries the records the projection read, the Lean source for every solved claim, and a manifest hashing every file. A document cannot contain its own hash, so the rendered document hash and the bundle hash are recorded in `MANIFEST.json` rather than here.

References

[1] Siemion Fajtlowicz, *Written on the Wall*, July 2004 compilation, 2004, https://independencenumber.wordpress.com/wp-content/uploads/2012/08/wow-july2004.pdf, located passages PDF page 47, Even definition; PDF page 82, Graffiti 322; content hash sha256:d2c779d2c28418b30ab1b4f84bf7112c0e000bca1f2d6a3c48d0baba78af7733.

[2] Milo Roucairol and Tristan Cazenave, *Refutation of Spectral Graph Theory Conjectures with Search Algorithms*, Communications in Computer and Information Science 2976, 2026, doi:10.1007/978-3-032-28263-7_2; arXiv:2409.18626v1, located passage Graffiti 322 discussion and C4 candidate; content hash sha256:fdda6abff9fd19f503a907f4170ea2040238cf0c0f0f317d280c2613fd17603f.

About this project[1]

[1] AdaIvy is an open-source AI-assisted mathematical research project that keeps retrieval, computation, formal checking, human review, and publication status as separate recorded boundaries. Source code and reproducibility materials: `https://github.com/jketts/AdaIvy`.